



FOOD FOR THOUGHT

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The hidden toll of stress on athletes and how to counter it

From national expectations to personal hurdles, **athletes face intense stress** that depletes vital nutrients, leading to emotional burnout and chronic fatigue.

Behind every gold medal and roaring crowd is a silent battle that few witness — one that tests an athlete's mind as much as their body. I've had the privilege of working with stars like Olympic medallists Kenneth Bednarek and P. V. Sindhu and cricketers like Shardul Thakur.

While they dazzle on the field, the real challenge lies in enduring the relentless pressure off it. From national expectations to personal hurdles, these athletes face intense stress that depletes vital nutrients, leading to emotional burnout and chronic fatigue.

Being a champion is more than just about physical strength — it's about surviving the mental grind.

The biochemistry of stress

When athletes face stress — whether physical or mental — their bodies go into fight-or-flight mode, releasing cortisol and adrenaline to boost energy, speed up heart rate, and sharpen focus. While this helps in the moment, it comes at a cost.

These stress hormones deplete key neurotransmitters like dopamine, serotonin, and acetylcholine, which regulate mood, focus, and muscle coordination. At the same time, stress increases the demand for essential nutrients like B vitamins, magnesium, calcium, and zinc.

Think of your body as a machine: neurotransmitters are the spark plugs, and vitamins and minerals are the fuel. When stress becomes constant, it's like running the

machine too hard — eventually, things break down. Without replenishing these critical nutrients, performance and overall health begin to suffer.

The impact of stress on the body

- **Dopamine:** Vital for motivation and focus, stress burns through dopamine quickly, leading to exhaustion and mental fatigue.
- **Serotonin:** The 'feel-good' chemical, serotonin, stabilises mood. Low levels due to stress cause anxiety, irritability, and even depression.
- **Acetylcholine:** Essential for muscle contraction and cognitive function, stress depletes this neurotransmitter, resulting in poor coordination, memory issues, and slower recovery.
- **B-complex vitamins:** Crucial for energy production, they get depleted by stress, leading to fatigue, slow recovery, and muscle cramps.
- **Magnesium:** Key for muscle relaxation and heart rhythm, stress